

MACHINE LEARNING ASSIGNMENT USING PYTHON

1. Introduction

Machine Learning is a branch of Artificial Intelligence that allows computers to learn from data and make predictions or decisions without being directly programmed for every task. In this practical work, a machine learning model was trained and evaluated using Python.

The purpose of this assignment is to demonstrate how a Python environment can be set up for machine learning, how a dataset can be loaded, how preprocessing can be performed, and how a model can be trained and evaluated. The practical also shows how the accuracy score and confusion matrix can be used to measure the performance of a machine learning model.

2. Objective of the Practical

The main objective of this practical is to set up a Python environment and use it to train and evaluate a machine learning model.

The specific objectives are:

1. To install or use a Python environment for machine learning.
2. To install the required Python libraries.
3. To prepare and load a dataset using pandas.
4. To perform basic preprocessing on the dataset.
5. To split the dataset into training and testing data.
6. To train a machine learning model.
7. To evaluate the model using accuracy score and confusion matrix.
8. To take screenshots of the code, output, accuracy score, and confusion matrix.

3. Tools Used

The following tools were used for this practical work:

- Python
- Google Colab / Jupyter Notebook
- pandas
- numpy
- scikit-learn
- matplotlib
- seaborn
- Microsoft Word

4. Setting Up the Python Environment

For this practical, Google Colab was used as the Python environment. Google Colab provides an online platform where Python code can be written and executed without installing Python manually on the computer.

The required Python libraries were installed using the following command:

```
pip install pandas numpy scikit-learn matplotlib seaborn
```

These libraries are important for data analysis, machine learning, and visualization.

Insert Screenshot 1: Python environment or Google Colab opened

Insert Screenshot 2: Installation of required Python libraries

5. Description of the Dataset

The dataset used for this practical is the Iris dataset. The Iris dataset is a common dataset used in machine learning for classification problems. It contains measurements of iris flowers and their corresponding flower classes.

The dataset contains the following input features:

- Sepal length
- Sepal width
- Petal length
- Petal width

The target/output classes are:

- Setosa
- Versicolor
- Virginica

The machine learning model uses the flower measurements to predict the correct flower class.

6. Loading the Dataset

The dataset was loaded using the pandas library. Pandas is used for handling and analyzing data in table form. After loading the dataset, the first few rows were displayed to check whether the data had been loaded correctly.

Insert Screenshot 3: Dataset loaded using pandas

7. Data Preprocessing

Data preprocessing is an important step in machine learning because it prepares the dataset for training. In this practical, the dataset was checked for missing values to ensure that there were no empty or invalid records.

The input features were separated from the target column. The feature values were then standardized so that all values would be on a similar scale. This helps the machine learning model to perform better.

Insert Screenshot 4: Dataset information and missing value check

8. Splitting the Dataset

The dataset was divided into training and testing sets. The training set was used to train the machine learning model, while the testing set was used to evaluate the performance of the model.

In this practical, 80% of the data was used for training and 20% was used for testing. This helps to check whether the model can make correct predictions on new data.

Insert Screenshot 5: Training and testing data split

9. Training the Machine Learning Model

A Logistic Regression model was used for the classification task. Logistic Regression is a supervised machine learning algorithm that can be used for classification problems.

The model was trained using the training dataset. During training, the model learned the relationship between the flower measurements and the flower classes.

Insert Screenshot 6: Model training completed successfully

10. Model Prediction

After training the model, predictions were made using the testing dataset. The predicted values were compared with the actual values in order to measure the performance of the model.

This step helps to determine whether the model was able to correctly classify the iris flowers.

Insert Screenshot 7: Predicted values

11. Model Evaluation

The model was evaluated using the accuracy score, classification report, and confusion matrix.

The accuracy score shows the percentage of correct predictions made by the model. A higher accuracy score means that the model performed well.

The classification report shows precision, recall, and F1-score for each class. These values help to understand how well the model performed for each flower type.

Insert Screenshot 8: Accuracy score

Insert Screenshot 9: Classification report

12. Confusion Matrix

A confusion matrix was used to show the number of correct and incorrect predictions made by the model. It compares the actual classes with the predicted classes.

The rows of the confusion matrix represent the actual classes, while the columns represent the predicted classes. Correct predictions appear along the diagonal of the matrix.

The confusion matrix was displayed using seaborn and matplotlib to make it easy to understand visually.

Insert Screenshot 10: Confusion matrix

13. Importance of the Practical

This practical is important because it shows the basic steps involved in building a machine learning model. It demonstrates how Python can be used to load data, preprocess data, train a model, and evaluate the result.

It also helps students understand how machine learning models make predictions based on data. The use of accuracy score and confusion matrix also shows how model performance can be measured.

14. Challenges Encountered

One challenge encountered during the practical was knowing where to add a new cell in Google Colab after installing the libraries. This was solved by using the  Code button to add a new code cell.

Another important point was making sure that each code cell was run in the correct order. If a cell is skipped, some variables may not be available for the next step.

15. Conclusion

In conclusion, this practical work demonstrated how to set up a Python environment for machine learning. The Iris dataset was loaded, preprocessed, split into training and testing sets, and used to train a Logistic Regression model.

The trained model was evaluated using accuracy score, classification report, and confusion matrix. The results showed how machine learning can be used to classify data and make predictions. This practical helped to improve understanding of the machine learning process using Python.